

## Unit 1 | Chapter 1: The Developer's World (Foundations)

| Learning Targets:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <ul style="list-style-type: none"> <li>❖ Technical: I can explain the mechanical differences between the Internet, the World Wide Web, and Intranets, and accurately trace the DNS (Domain Name System) lookup process. (NV 6.1.1, 6.1.2)</li> <li>❖ Technical: I can decode a URL, identify the role of SSL/HTTPS, and differentiate between web hosting models and Content Delivery Networks (CDNs). (NV 5.6.1, 6.2.1)</li> <li>❖ Technical: I can identify the three core web languages (HTML, CSS, JS) and evaluate foundational UI/UX design principles like visual hierarchy and white space. (NV 4.3.1, 5.1, 5.2)</li> <li>❖ Employability: I can demonstrate professional decorum by identifying developer career pathways, utilizing CTSOs (SkillsUSA/FBLA) to build a portfolio, and participating in parliamentary procedure. (NV 1.1, 1.2, WRS 1.1.4)</li> <li>❖ Employability: I can demonstrate probity and cultural competency by strictly adhering to intellectual property laws (Copyright/Creative Commons), web accessibility mandates, and secure file naming conventions (e.g., index.html). (NV 2.1, WRS 1.1.2, 1.1.5)</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Compelling Question:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| How do the mechanical foundations of the internet, combined with strict professional ethics and communication skills, empower a developer to architect the modern web?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Score                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p><b>I can:</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Synthesize internet infrastructure concepts to architect a scalable hosting solution that prioritizes security and elasticity for a viral application, demonstrating elite analytical reasoning. (NV 5.6.1, WRS 1.2.3)</li> <li><input type="checkbox"/> Formulate a comprehensive digital ethics and accessibility policy for a corporate development team that strictly adheres to copyright laws and WCAG standards, demonstrating profound probity and cultural competency. (NV 2.1, WRS 1.1.2, 1.1.5)</li> <li><input type="checkbox"/> Lead a simulated CTSO chapter meeting using flawless parliamentary procedure to systematically resolve a team conflict. (NV 1.2, WRS 1.1.4)</li> </ul> |
| 3.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | In addition to score 3.0 performance, partial success at score 4.0 content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p><b>I can:</b></p> <ul style="list-style-type: none"> <li><input type="checkbox"/> Execute the DNS lookup sequence and explain the Client-Server model in detail. (NV 6.1.2)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|            | <input type="checkbox"/> Apply the rules of copyright, creative commons, and secure file naming (e.g., all lowercase, no spaces, index.html). (NV 2.1)<br><input type="checkbox"/> Define the distinct roles of HTML (Structure), CSS (Style), and JavaScript (Logic). (NV 5.1, 5.2)<br><input type="checkbox"/> Explain the value of CTSOs, soft skills (like Rubber Duck Debugging), and the Agile "Project Triangle" in advancing a tech career. (NV 1.1, 1.4)<br><input type="checkbox"/> Demonstrate diligence and intrinsic motivation by completing foundational web theory assignments on schedule. (WRS 1.1.1) |
| <b>2.5</b> | No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>2.0</b> | <b>I can</b><br><input type="checkbox"/> Define common terminology and identify syntax: (WEB.4.2.1) <ul style="list-style-type: none"> <li><input type="checkbox"/> Internet vs. Web, IP Address, Domain Name, URL, HTTPS, DNS, HTML/CSS/JS, CTSO, SkillsUSA, FBLA, Parli Pro, Copyright, MVP, Soft Skills, Visual Hierarchy.</li> </ul> <input type="checkbox"/> Identify the importance of punctuality (WRS 1.1.1), probity and ethical adherence (WRS 1.1.2), professional decorum in team communication (WRS 1.1.4), inclusive collaboration (WRS 1.1.5), and analytical reasoning (WRS 1.2.3).                     |
| <b>1.5</b> | Partial success at score 2.0 content, and major errors or omissions at 3.0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>1.0</b> | With help, partial success at score 2.0 and score 3.0 content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>0.5</b> | With help, partial success at score 2.0 but not at score 3.0 content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>0.0</b> | Even with help, no success or attempt made.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |